Comparative Time-Series Forecasting of Food Inflation Using El Niño Indices: ARIMA vs. Prophet vs. LSTM

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***Abstract*—Climate variability may influence agricultural conditions and food prices, but whether a global climate indicator improves a specific food-inflation forecasting model is an empirical question. This study evaluates the predictive contribution of the Oceanic Niño Index (ONI) to monthly Indian food inflation using 168 aligned observations from January 2012 to December 2025. Four implemented forecasting configurations are compared: an autoregressive integrated moving average model with an exogenous ONI regressor (ARIMAX), Prophet with an ONI regressor, a multivariate long short-term memory (LSTM) network using food inflation and ONI, and an inflation-only LSTM baseline. Reported predictions are evaluated over January 2024–December 2025 with mean absolute error, root mean squared error, symmetric mean absolute percentage error, and mean absolute scaled error. The historical-only LSTM obtains the smallest reported errors (MAE 1.3335; RMSE 1.6460), followed by LSTM with ONI (MAE 1.5245), ARIMAX (4.6550), and Prophet (5.0169). Adding ONI therefore does not improve the LSTM baseline in this benchmark. However, forecasting information and horizons differ across model families, so the rankings should not be interpreted as a controlled estimate of ONI's causal contribution or universal predictive value. The study illustrates the value of transparent baseline comparisons and identifies requirements for stricter future evaluation.**

***Index Terms*—Food inflation, El Niño, Oceanic Niño Index, time-series forecasting, LSTM, ARIMAX, Prophet.**

I. Introduction

Food-price inflation is relevant to household purchasing power, agricultural supply chains, and economic monitoring. In India, the Consumer Food Price Index (CFPI) tracks changes in retail prices of food items consumed by households [1]. Food inflation depends on several interacting influences, including agricultural production, rainfall, transportation, seasonality, market conditions, and policy responses. A forecast obtained from past inflation alone may therefore omit potentially useful external signals.

El Niño is associated with warmer-than-normal sea-surface conditions in the equatorial Pacific, within the broader El Niño–Southern Oscillation (ENSO) system. The Oceanic Niño Index (ONI) summarizes three-month mean sea-surface temperature anomalies in the Niño 3.4 region and provides a standardized measure of this climate variation [2]. A plausible connection between climate conditions and agricultural supply motivates considering ONI as an auxiliary predictor, but that physical plausibility does not ensure improved prediction of national food inflation. Recent research on India documents heterogeneous and delayed food-inflation responses to distinct climate shocks; the study addresses different climate variables, frequencies, and research questions from those analyzed here [3].

Research question: Does including ONI improve reported out-of-sample prediction accuracy relative to a model using only historical food inflation? To investigate this question, the present work combines monthly food-inflation and ONI series, constructs four forecasting configurations, and compares their saved predictions over a common 24-month calendar window. The ARIMA-family member is implemented specifically as ARIMAX: an ARIMA model incorporating an exogenous ONI-based regressor.

The contributions are threefold. First, the project evaluates both climate-augmented and inflation-only recurrent models rather than assuming the climate feature is beneficial. Second, it records four complementary forecast-error measures and examines the behaviour of model predictions across the holdout period. Third, it exposes processed datasets and stored outputs through a FastAPI service and a React-based analysis dashboard. Its principal finding is a *negative predictive result* for the ONI addition in this configuration; it is not a finding that ENSO does not influence Indian food prices.

II. Related Work and Research Gap

A. Statistical and Decomposable Forecasting

Autoregressive integrated moving average models describe temporal persistence and changes in a series. When external variables enter the model, the ARIMAX formulation can test whether those regressors are useful in a fitted predictive equation. The implementation uses the SARIMAX state-space functionality of `statsmodels` [4]. Prophet models trend, seasonality, and possible external regressors in a decomposable forecasting framework; Taylor and Letham introduced the approach as a practical forecasting method for time series with recurring patterns [5].

These approaches are useful reference models, but their ability to extrapolate through structural changes depends on their specification, assumptions, and forecast horizon. Supplying contemporaneously constructed external regressors during a held-out period also creates an information-timing question: values known retrospectively are not automatically available at the date a genuine future forecast would have been issued.

B. Sequence Models and Forecast Evaluation

LSTM networks were proposed to learn dependencies over time while mitigating limitations of conventional recurrent networks [6]. The implemented recurrent layers use the documented Keras LSTM interface [7]. A multivariate LSTM can combine preceding inflation observations with preceding ONI observations. Nevertheless, additional predictors can worsen generalization when the sample is limited, the relationship is weak or unstable, or model complexity is not matched to the signal.

Forecasting studies also require clear evaluation rules. Hyndman and Koehler analyze shortcomings of several percentage-based error measures and motivate mean absolute scaled error (MASE) for interpretable comparisons [8]. Formal comparative forecast inference requires care about dependent errors and common evaluation conditions, as highlighted by Diebold and Mariano [9]. The present study reports descriptive error rankings, *not* a hypothesis test that one model significantly outperforms another.

C. Gap Addressed by This Study

The specific gap addressed is empirical rather than a claim to invent a new forecasting algorithm: does an ONI feature improve predictive accuracy for the assembled monthly Indian food-inflation series compared with a historical-only LSTM? The matched baseline is essential because an ONI-augmented model's reasonable absolute performance does not establish that ONI supplied incremental information. A second goal is to describe limitations of the available benchmark rather than overgeneralize its results.

III. Data and Preprocessing

A. Sources and Variables

The project contains raw CPI/CFPI-related records and a NOAA monthly ONI time series. Data preparation produces `cpi_oni.csv`, with 168 monthly rows from January 2012 through December 2025. Each processed row records `date`, `food_inflation`, `oni`, and `month`. The first is a monthly timestamp, the second a reported or derived year-on-year inflation rate in percent, and the third a temperature-anomaly index rather than an inflation percentage.

A crucial distinction is that ONI measures a climate anomaly; it is not a local Indian rainfall measurement, crop-yield estimate, or economic policy variable. The CPI/CFPI records constitute the target-source data, and the external climate series is supplied by NOAA's Physical Sciences Laboratory [2]. Any changing data definitions or source revisions would need to be reconciled in a full operational study.

B. Preparation Pipeline

The preprocessing notebook converts CPI year/month fields to a first-of-month date and sorts records in chronological order. Where a source inflation value is missing and appropriate index history exists within the same CPI series grouping, year-on-year inflation is estimated by

$$\pi_t = 100\left(\frac{I_t}{I_{t-12}} - 1\right), \tag{1}$$

where I_t denotes the corresponding price index. The ONI file is separately date-parsed and its missing-value sentinels are replaced with missing values. The two series are then joined by month; the resulting timeline is inspected for duplicates and missing dates before being saved.

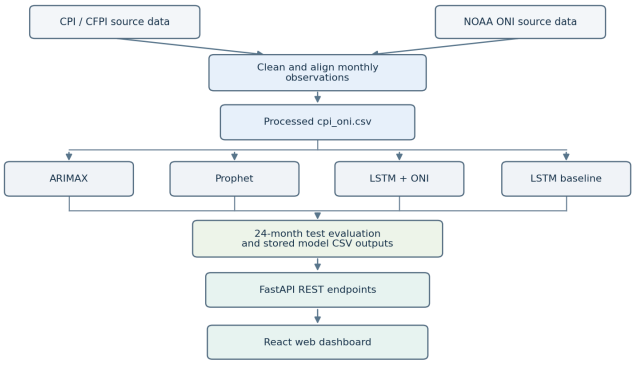


Fig. 1. Implemented project workflow: source preprocessing, four models, common dated evaluation, and presentation of stored outputs through the API and dashboard.

The common chronological split consists of 144 training months (January 2012–December 2023) and 24 holdout months (January 2024–December 2025). The final reported evaluation uses the saved prediction outputs produced by the individual experiment notebooks; values were not recomputed by a live dashboard request.

C. Input Features and Temporal Context

To reduce inadvertent leakage, the LSTM preprocessing code fits scikit-learn `MinMaxScaler` instances only on training observations and applies those transformations to validation/test inputs [10]. A temporal input window uses only the preceding L months for the current prediction target. The baseline model consumes historical inflation values; the ONI-augmented model consumes paired historical inflation and ONI values. The tested window lengths are not identical and are stated explicitly below.

IV. Forecasting Methods

A. ARIMAX with Lagged ONI

The implemented ARIMA-family model uses the `SARIMAX` estimator with nonseasonal order $(p, d, q) = (0, 1, 3)$, a constant term, and an ONI-derived external regressor. The regressor is a lagged three-month average,

$$x_t = \frac{\mathrm{ONI}_{t-1} + \mathrm{ONI}_{t-2} + \mathrm{ONI}_{t-3}}{3}. \tag{2}$$

This design makes the predictor depend on earlier ONI observations rather than the current month's ONI value. The

TABLE I
FINAL IMPLEMENTED FORECASTING CONFIGURATIONS

Model	Target inputs/structure	Test prediction setup
ARIMAX	Inflation + lagged ONI MA3; $(0, 1, 3)$	Multi-step; test ONI values supplied
Prophet	Annual seasonality + lagged ONI MA3	Test ONI values supplied
LSTM + ONI	Two features; 18-month history	Rolling observed history
LSTM baseline	Inflation only; 12-month history	Rolling observed history

model is fitted to pre-2024 training data, and its stored 2024–2025 predictions are generated through a multi-step forecast using exogenous regressor values supplied for the held-out period.

B. Prophet with Climate Regressor

The Prophet notebook enables annual seasonality and uses additive regression for the same three-month lagged ONI feature in Eq. (2). The final configuration sets `changepoint_prior_scale`=0.10, `seasonality_prior_scale`=1.0, and `interval_width`=0.95. The model produces point predictions alongside lower and upper prediction bounds. Feature and hyperparameter experiments in the notebook use an earlier validation period; the final test output is the January 2024–December 2025 sequence. The quoted point-error metrics assess the central prediction, not the calibration of the interval bounds.

C. LSTM with ONI

The multivariate LSTM is trained on a two-feature sequence (food inflation, ONI) with an 18-month lookback. Its architecture is an LSTM layer of 64 units, dropout probability 0.2, a dense layer of 32 rectified-linear units, and a one-neuron regression output. The optimizer is Adam and the training loss is mean squared error. A training-only validation subset is used to choose an epoch count; the final model is then fitted to the training portion and applied to 24 dated test samples.

D. Inflation-Only LSTM Baseline

The baseline has the same general layer sequence but receives only preceding food inflation and uses a 12-month lookback. It is intentionally retained as the crucial negative-control style comparison: if ONI contains useful incremental forecasting information for the chosen setup, the ONI model should outperform a sufficiently competent inflation-only baseline. However, the unequal 12- and 18-month windows mean this is not a strict single-variable ablation.

V. EXPERIMENTAL EVALUATION

A. Evaluation Setup

All four final prediction files contain dated results for the same 24 holdout months, and their actual inflation values were checked for alignment in the model-comparison notebook. The notebook calculates four point-forecast error measures. For observations y_t and predictions $\hat{y}_t$, with $n = 24$,

$$\mathrm{MAE} = \frac{1}{n}\sum_{t=1}^{n}|y_t - \hat{y}_t|, \tag{3}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{t=1}^{n}(y_t - \hat{y}_t)^2}, \tag{4}$$

$$\mathrm{sMAPE} = \frac{100}{n}\sum_{t=1}^{n}\frac{2|y_t - \hat{y}_t|}{|y_t| + |\hat{y}_t| + \varepsilon}, \tag{5}$$

$$\mathrm{MASE} = \frac{\mathrm{MAE}}{\frac{1}{T-1}\sum_{t=2}^{T}|y_t^{\mathrm{tr}} - y_{t-1}^{\mathrm{tr}}|}, \tag{6}$$

where y_t^{tr} is a training observation, $T = 144$, and $\varepsilon = 10^{-8}$ for numerical stability. MAE and RMSE are measured in percentage points because the target is an inflation rate. A MASE below one indicates an error below the in-sample naive-change scale used in the denominator, not necessarily superior performance to every out-of-sample naive forecast.

The notebook ranks models separately on each metric (smaller is better), takes the mean rank, and assigns a final ordering. The recorded ranking is the same for all four measures. No confidence intervals, repeated-run distributions, or paired statistical significance tests for relative model performance are reported.

B. Important Information-Set Distinction

The shared calendar window should not be confused with a fully matched forecasting protocol. In the two LSTM notebooks, test windows are constructed from the final training observations and the observed test sequence. Later test-month inputs therefore incorporate earlier *observed* test-month inflation values. The setup resembles sequential one-step conditional prediction rather than a single uninterrupted 24-month forecast issued in December 2023. In contrast, the ARIMAX code generates a multi-step prediction path while supplying lagged ONI exogenous values from the test calendar. Prophet similarly receives test-date ONI-derived regressors.

Accordingly, these results support comparisons of the *implemented prediction procedures*, not a definitive ranking under equal forecast-origin, horizon, and information availability conditions. This distinction is central to responsible interpretation of the relative errors.

VI. RESULTS AND DISCUSSION

A. Reported Overall Performance

Table II reports the final values saved in the model-comparison output, rounded to four decimal places. The inflation-only LSTM ranks first on all metrics: MAE 1.3335, RMSE 1.6460, sMAPE 43.8900%, and MASE 0.9910. The LSTM with ONI comes second with MAE 1.5245, followed by ARIMAX (4.6550) and Prophet (5.0169).

The ONI-augmented LSTM's MAE is approximately 14.33% higher than the baseline MAE, a relative difference computed as $(1.5245/1.3335 - 1) \times 100$ using the full-precision stored

TABLE II
REPORTED TEST ERRORS, JANUARY 2024–DECEMBER 2025

Model	MAE	RMSE	sMAPE (%)	MASE	Rank
LSTM baseline	**1.3335**	**1.6460**	**43.8900**	**0.9910**	**1**
LSTM + ONI	1.5245	1.9422	49.6371	1.1330	2
ARIMAX	4.6550	6.3055	85.4437	3.4594	3
Prophet	5.0169	6.6850	87.4038	3.7284	4

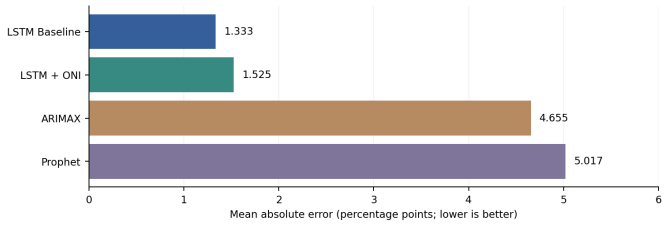


Fig. 2. Mean absolute error by model, reproduced from the finalized project report's saved comparison results; lower is better.

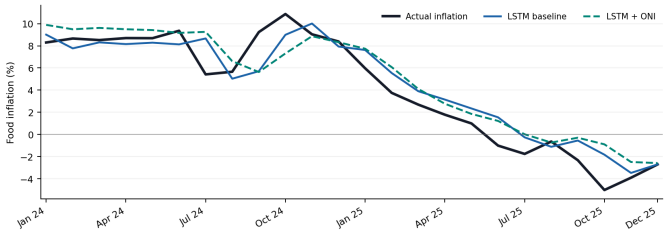


Fig. 3. Actual food inflation and predictions from the two LSTM configurations on the common test calendar (reproduced from the project report).

metrics. Because the ONI model also uses a different history length, this gap does not isolate the effect of ONI alone. Nevertheless, it provides no evidence that the tested ONI-augmented configuration improves the baseline on this holdout window. The two statistical procedures report substantially higher errors.

B. Temporal Behaviour

Figure 3 displays the project report's actual-versus-predicted chart for the two recurrent configurations. Both LSTM prediction sequences broadly follow the later decline in inflation, although month-to-month deviations remain and errors are not negligible. In contrast, the ARIMAX and Prophet stored paths remain largely positive while observed inflation moves down and eventually becomes negative in parts of 2025 (Fig. 4). The widening differences contribute to their relatively high MAE and RMSE values on this particular historical episode.

The sMAPE values are relatively large despite the lower LSTM errors. This measure is sensitive when the actual and/or forecast value approaches zero and can be difficult to interpret for signed inflation rates. Consequently, MAE, RMSE, and MASE are more informative when read alongside temporal plots and the target units; no single metric should be treated as an exhaustive indicator of forecast quality [8].

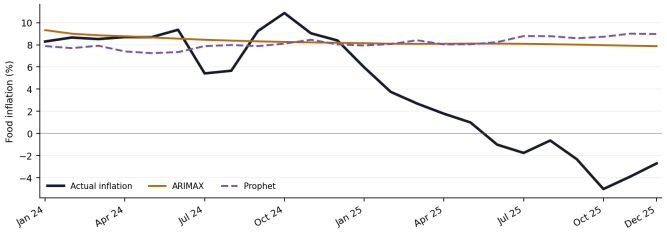


Fig. 4. Actual food inflation compared with the ARIMAX and Prophet test-period prediction paths (reproduced from the project report).

C. What the ONI Result Means

An ONI feature is theoretically relevant to climate variability, yet adding it does not guarantee improved inflation forecasts. Possible explanations include the indirect and delayed path from Pacific temperature anomalies to local agriculture and prices, strong inflation persistence, variation in feature lag, finite training sample size, and specification or training effects. These explanations are *hypotheses*, not mechanisms established by the current experiments. The results cannot tell whether other ONI lags, rainfall measures, regional food categories, or comparable model setups would lead to different outcomes.

Likewise, a difference in errors is not evidence of an economically significant or statistically significant improvement without uncertainty analysis under an appropriate common evaluation design. The present results should be reported as descriptive rankings on the saved test series, not as proof of a causal relationship or its absence.

VII. System Implementation and Reproducibility

The project uses offline Jupyter notebooks to explore source data, prepare the combined monthly dataset, fit candidate models, save dated prediction tables, and calculate metrics. The deployment architecture separates these experimental computations from presentation. A Python FastAPI backend reads saved CSV files and exposes endpoints including `/data`, `/predictions`, `/comparison`, and `/forecast`; a React/Vite dashboard with Recharts requests and visualizes those outputs.

The live API serves stored results. Its `/forecast` endpoints return the latest dated records present in the prediction CSV files and do *not* imply that models are retrained or that current future-month forecasts are generated on request. The dashboard permits inspection and comparison of evaluated outputs; it is not an independently validated operational inflation forecasting system.

For reproducibility, the codebase contains the data-processing, model-development, and evaluation notebooks and saved results. However, faithful reconstruction requires preservation of data snapshots, software versions, neural-network training conditions, and any random-state controls. The available evidence does not establish identical results from independent reruns or provide trained-model artifacts for all deployed inference scenarios. The numerical findings in this paper are therefore described as *repository-reported, saved test*

results rather than fresh experiments conducted for manuscript preparation.

VIII. Limitations and Further Research

The central limitation is that prediction horizons and information availability differ across model families. For a stronger incremental-value study, both LSTM variants should use matched lookback lengths, architecture, feature timing, validation strategy, and training budgets, differing only in the presence or absence of ONI. All models should be evaluated through a rolling-origin backtest with a specified issuance date, a common h-step horizon, and a rule that allows only external information realistically available at that issuance time.

Further work should compare simple persistence, seasonal-naive, and autoregressive controls with the neural methods; evaluate alternative ONI lag structures and weather covariates; examine residual autocorrelation and uncertainty calibration; and report variability across multiple neural training runs. Regional and food-category-level analyses could better reflect heterogeneous climate exposure. Where the design permits, formal forecast-comparison procedures and confidence intervals should be considered. Any operational deployment would additionally require scheduled data updates, provenance tracking, external-feature forecasts, monitoring, and prospective validation.

IX. Threats to Validity and Interpretation

A. Internal and Construct Validity

The experimental feature comparison is not a controlled ablation because the baseline and ONI-augmented LSTMs use 12- and 18-month history windows, respectively. Even if both networks share similar layers, differences in temporal context, validation trajectories, and learned weights can change the error independently of the ONI feature. Consequently, the observed MAE increase of 14.33% cannot be attributed entirely to ONI. The correct conclusion is that the *implemented ONI-augmented configuration* fails to outperform the *implemented inflation-only configuration* on the reported test data.

A second internal-validity threat arises from the timing of information supplied to different procedures. A model that is handed observations from earlier months within the test period solves a different task from one asked to forecast the complete two-year sequence from a fixed origin. Similarly, lagged climate regressors computed from observed future-period ONI become available retrospectively during evaluation but may not be fully known for a future-dated real-time forecast. Thus the shared 24-month test calendar is a necessary consistency check, but it is not sufficient for matching the information sets.

The construct of "climate-informed forecasting" is also narrower here than the mechanisms linking climate to food markets. ONI is a large-scale sea-surface temperature anomaly; its relationship to Indian agricultural output and retail food prices may depend on rainfall, local cropping patterns, trade, storage and policy responses. A model with only a short ONI history is not an exhaustive test of climate features. Neither a negative prediction result nor a low error from an autoregressive model can establish the presence or absence of a causal ENSO effect.

TABLE III
Requirements for a Stronger ONI Incremental-Value Test

Evaluation dimension	Proposed matched condition
Forecast origin	Rolling-origin predictions at identical timestamps
Prediction horizon	Same h months ahead for every candidate
Feature availability	Only inflation and ONI observed/forecast by the origin
Controlled LSTM	Same lookback, layers, tuning budget; vary ONI only
Uncertainty	Multiple seeds and uncertainty estimates for error differences
Reference baselines	Persistence, seasonal naive, and simple autoregressive models

B. Data and External Validity

The processed series contains 168 monthly observations, a small sample relative to the parameterization of neural models. The training data cover particular economic and climate episodes, and the holdout includes only January 2024–December 2025. A model ranking from this interval may change during other inflation regimes, alternative base-year revisions, or different geographic and commodity aggregations. The CPI preprocessing also fills some missing inflation entries from a 12-month percentage change of a price index; full replication requires careful validation of the source grouping and consistency of index definitions.

The reported numerical values come from saved notebook outputs rather than independent repeated experiments. Neural initialization, dependency versions and model selection can introduce variability. The outputs do not establish the distribution of error differences over repeated seeds, and no calibration or interval-coverage analysis is reported for probabilistic claims. The Prophet prediction file includes upper and lower bounds, but its central forecast alone is used in the four-metric ranking.

C. A Confirmatory Evaluation Protocol

A follow-up study designed to answer the incremental-value question more conclusively should (i) define a single forecast horizon, or report several horizons separately; (ii) use the same rolling prediction origins for every method; (iii) freeze feature information at each origin; (iv) compare identical LSTM architectures and lookback lengths with versus without ONI; (v) reserve the outer test data exclusively for evaluation; and (vi) quantify run-to-run uncertainty and test-error variability. If suitable, paired forecast-loss comparisons could then be evaluated using methods discussed by Diebold and Mariano [9].

This proposed protocol is *future work*; it has not been run as part of the reported benchmark. Its purpose is to distinguish descriptive model comparison from a controlled claim about how much ONI improves the forecasts. Such caution is especially important for food inflation: the numerical outputs are research prototypes and should not be used as

an unvalidated basis for household procurement, commodity trading, or policy decisions.

X. Conclusion

A monthly Indian food-inflation forecasting benchmark was implemented using 168 historical records, an ONI climate series, and four model configurations. Over the recorded January 2024–December 2025 test window, the inflation-only LSTM reports the lowest errors on all four evaluation metrics. The ONI-augmented LSTM ranks second and does not surpass that baseline. ARIMAX and Prophet have larger errors in their saved outputs during a period that includes a substantial decline in observed inflation.

These results answer the present project's operational research question with an important qualification: ONI did not improve the tested LSTM configuration, but unequal lookback lengths and different forecast information conditions prevent isolating the climate feature's standalone contribution across all methods. The work supports transparent reporting, disciplined baseline selection, and a stricter future evaluation protocol; it does not establish that El Niño has no influence on Indian food inflation.

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